Cloud Computing Fall 2026

Assignment No.1

Submitted To: Dr. Muhammad Abdullah

Submitted By: Muhammad Imran Mian

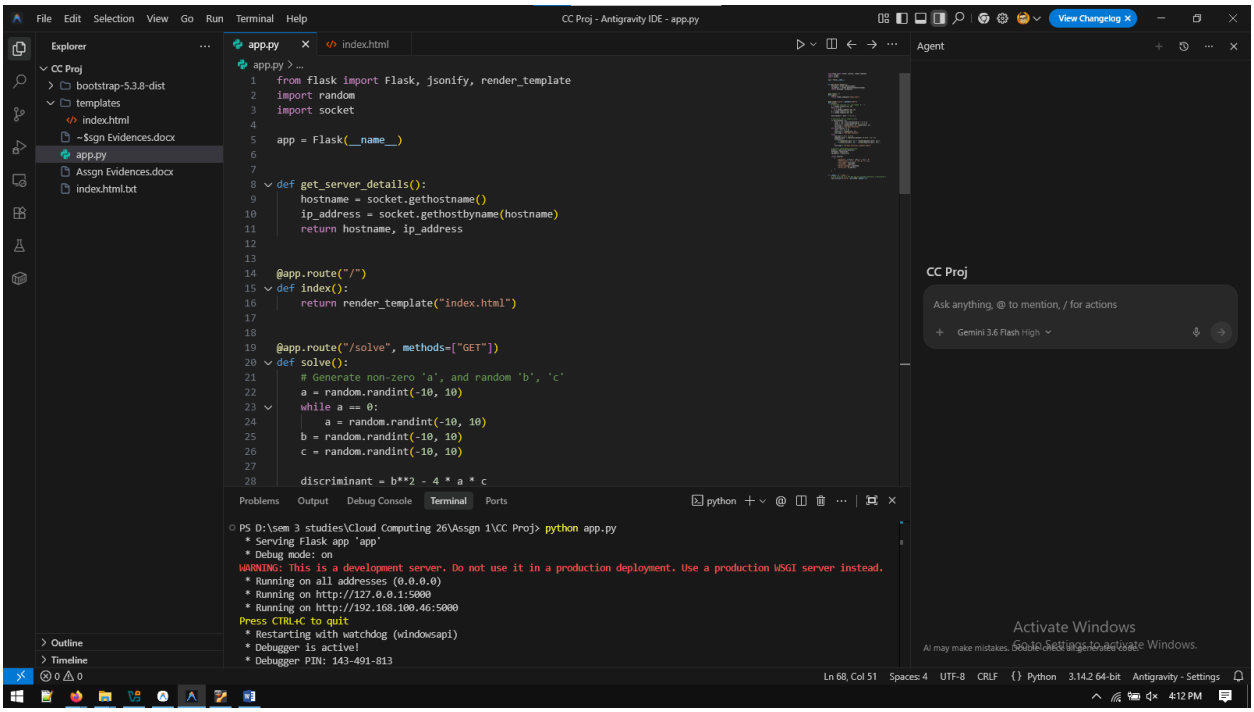
Roll No.: MSDSF25-A007

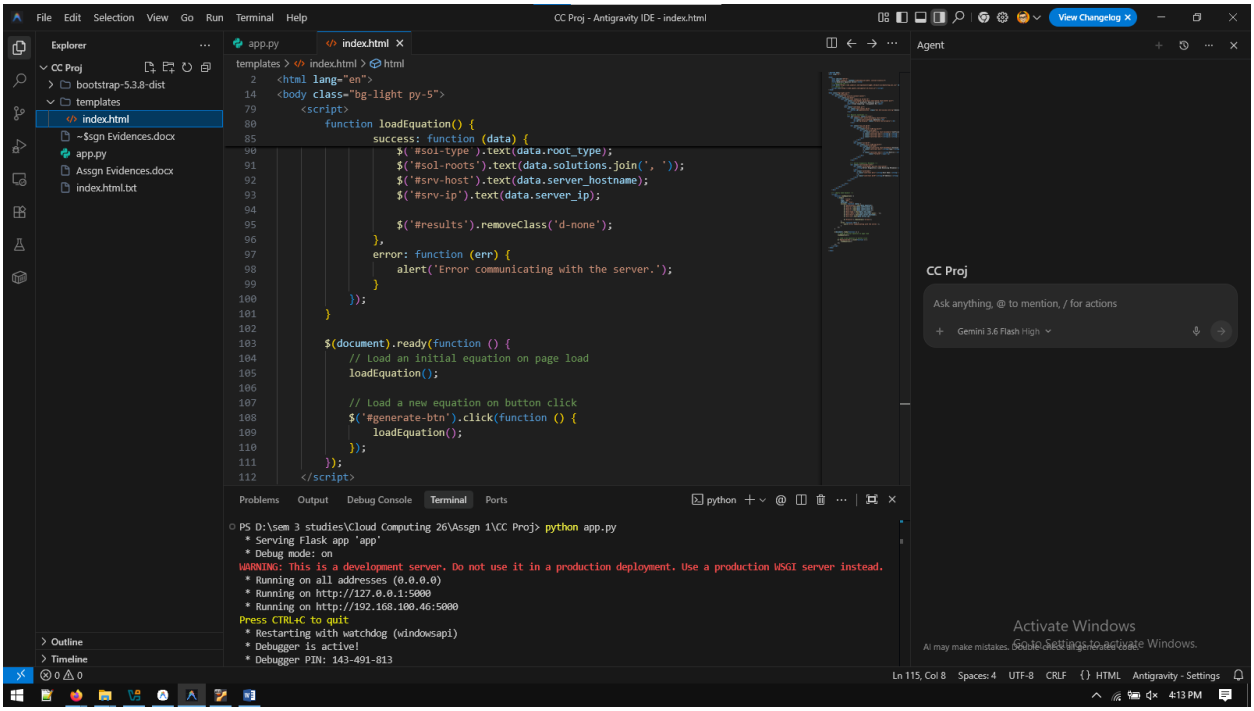
Dated: 28-09-2026

Milestone 01: Develop a Sample Web Application

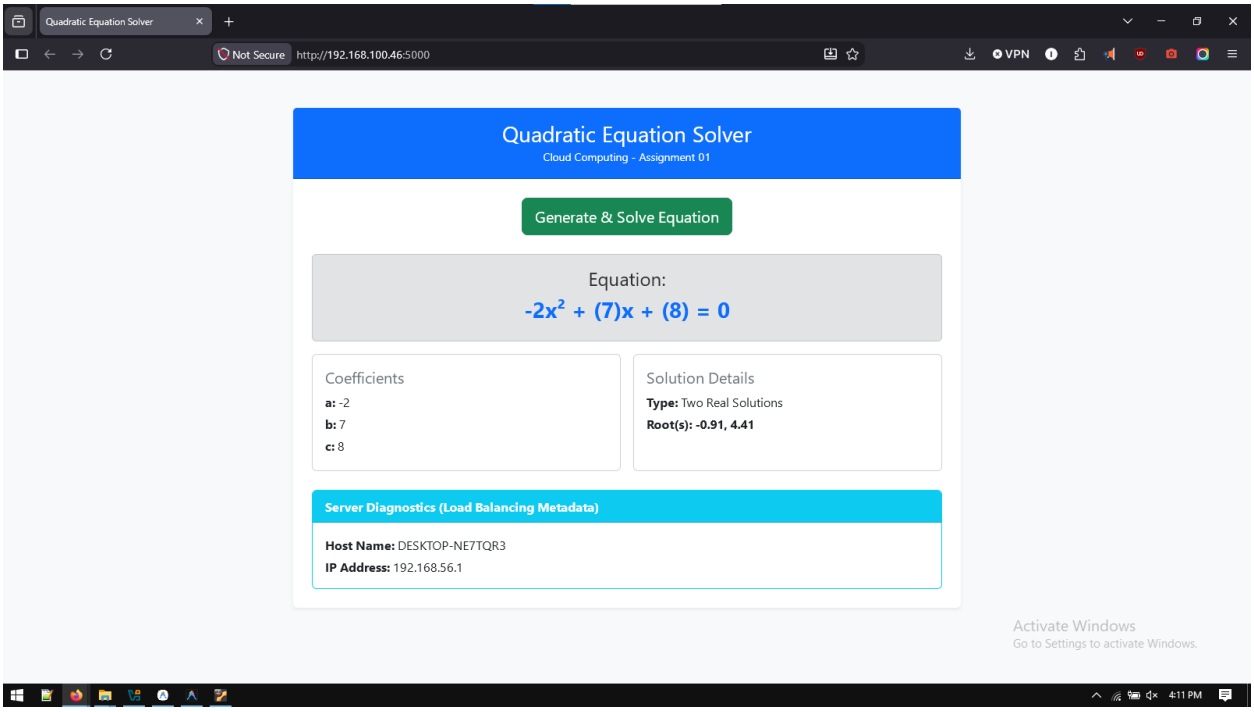
Backend Python (BE Logic) – Flask Framework (Web Routing

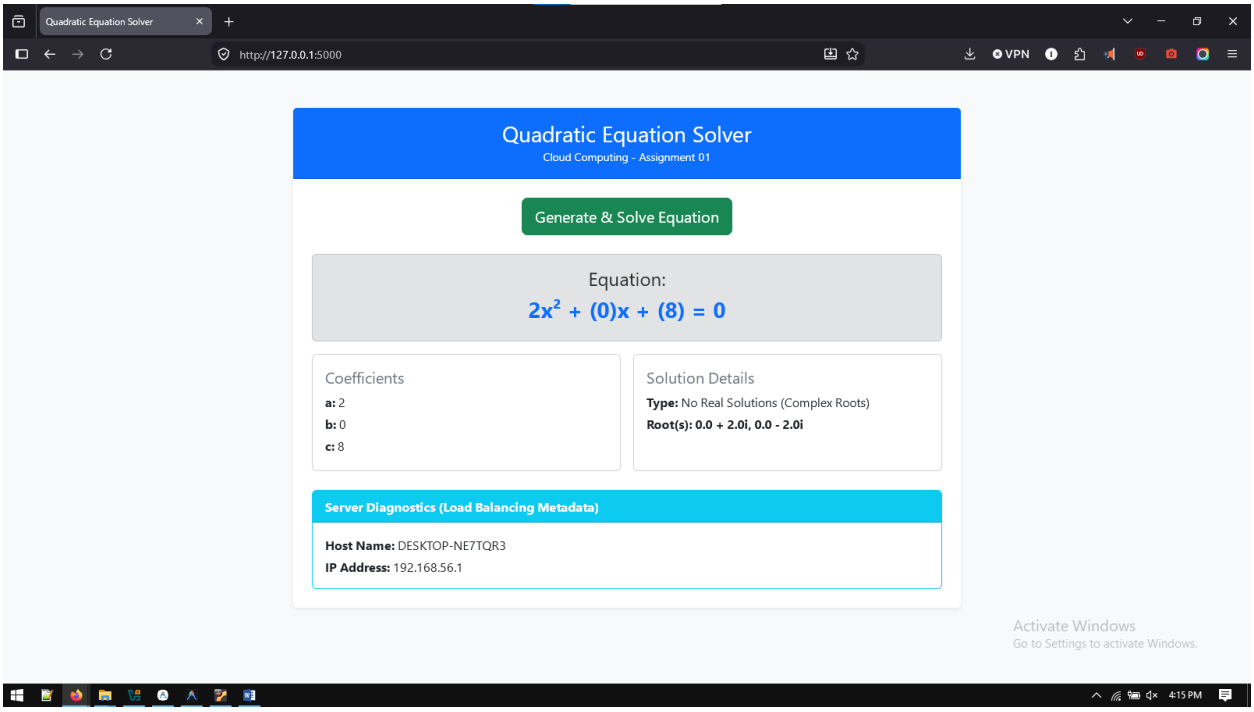
Frontend: HTML5 (Page Structure), JavaScript (Interactivity), JQuery (DOM Manipulation), AJAX (User Interaction)





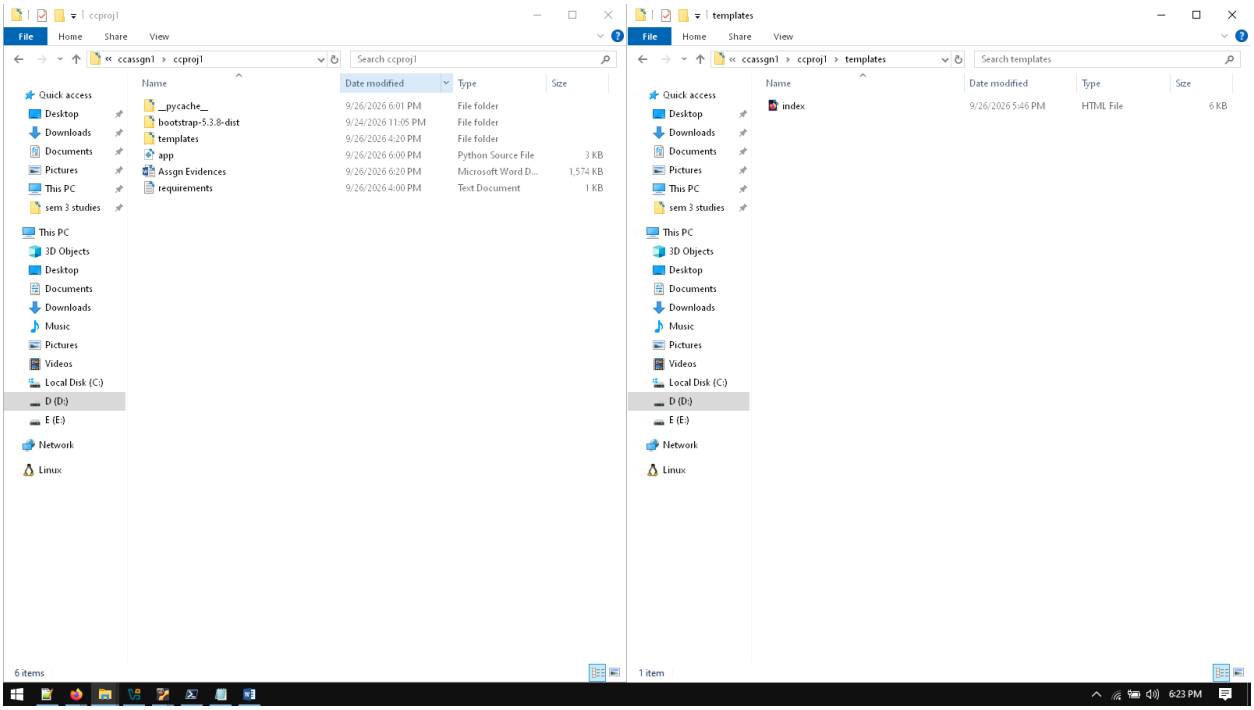
➔ Application running on HOST OS



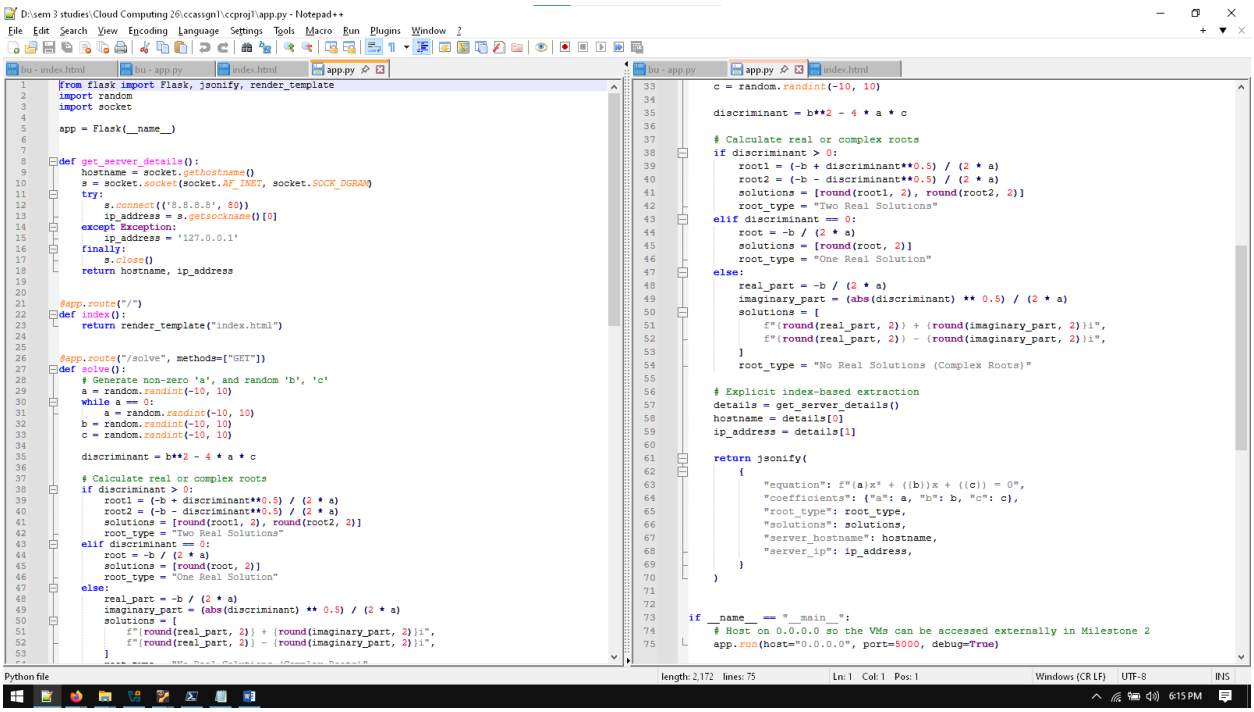


Milestone 02: Host your Web Application on Ubuntu Virtual Machines

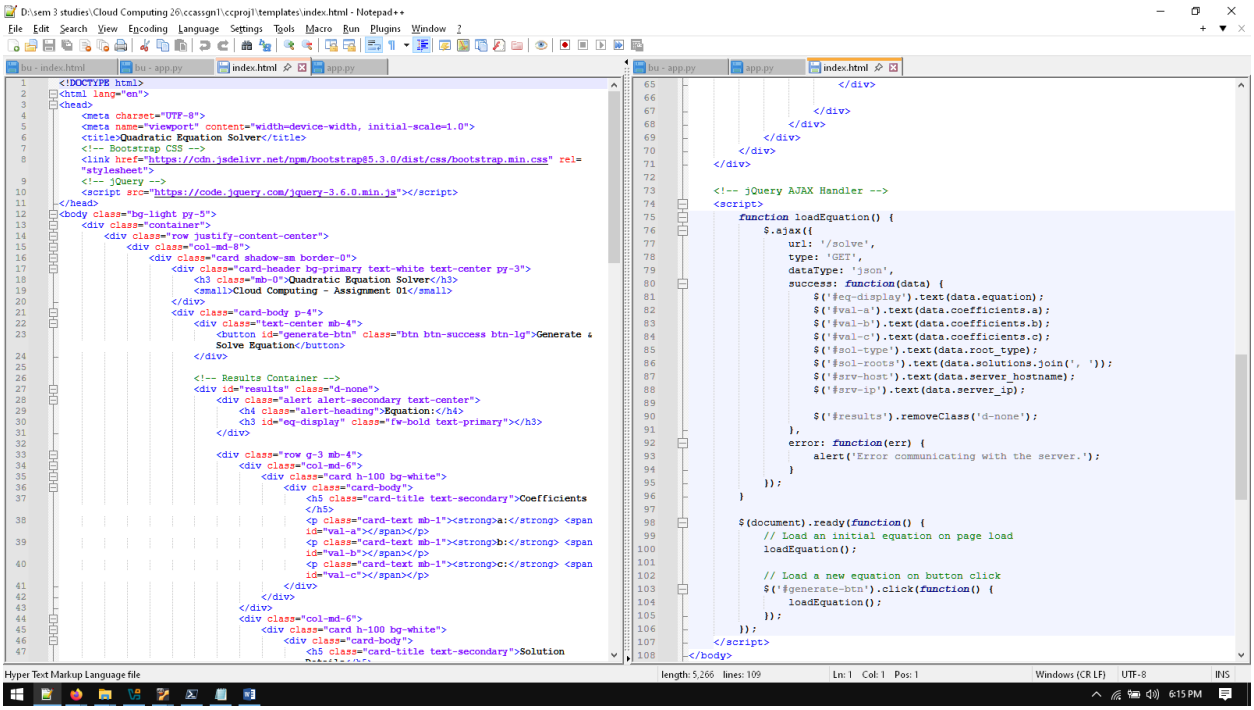
➔ Project files



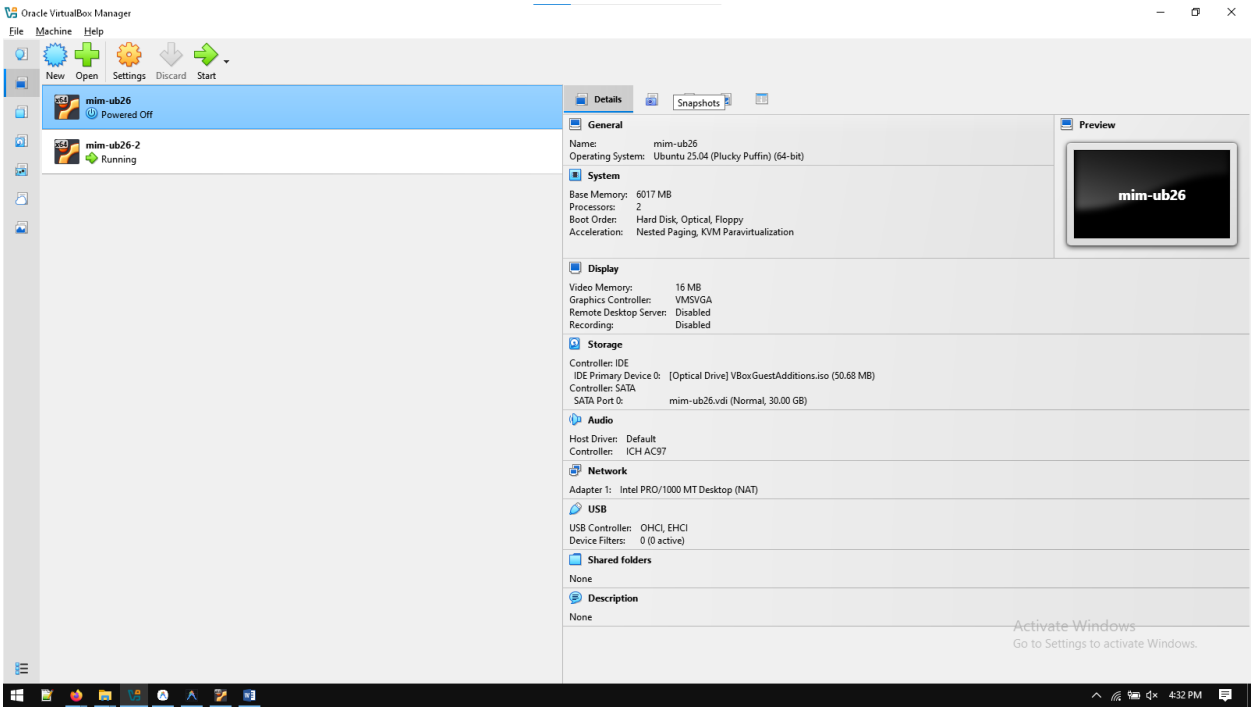
➔ App.py script

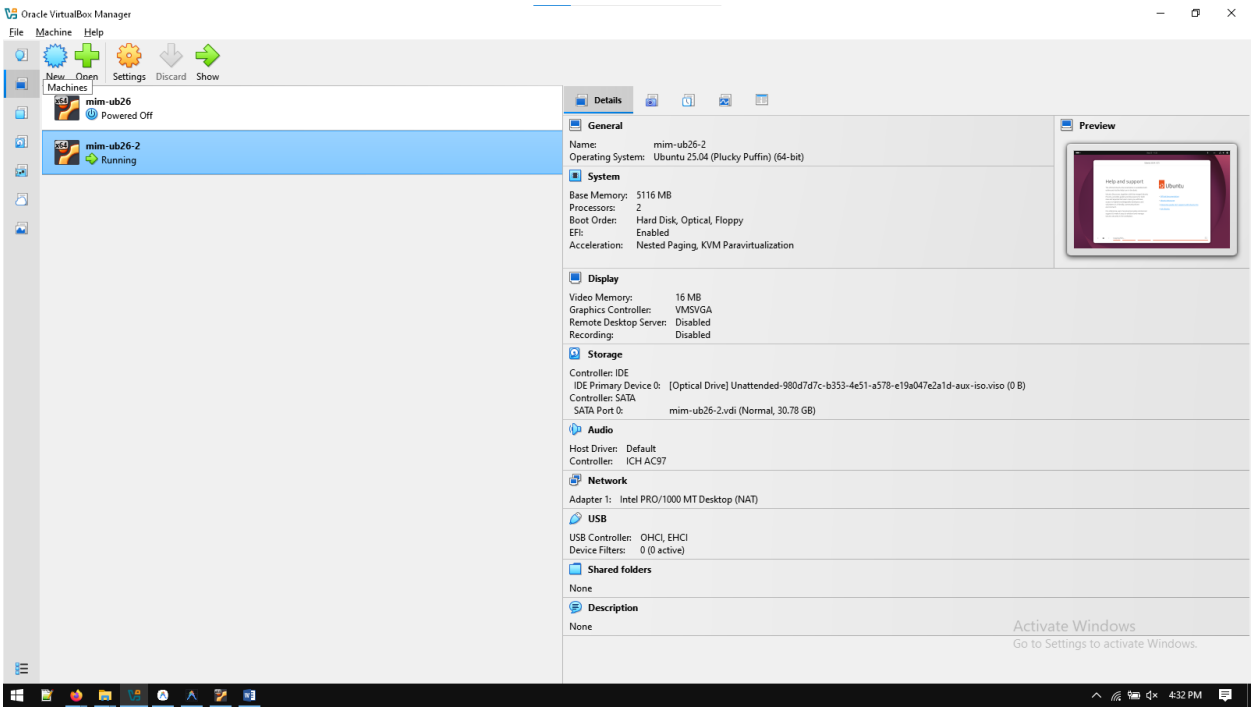


➔ Index.html



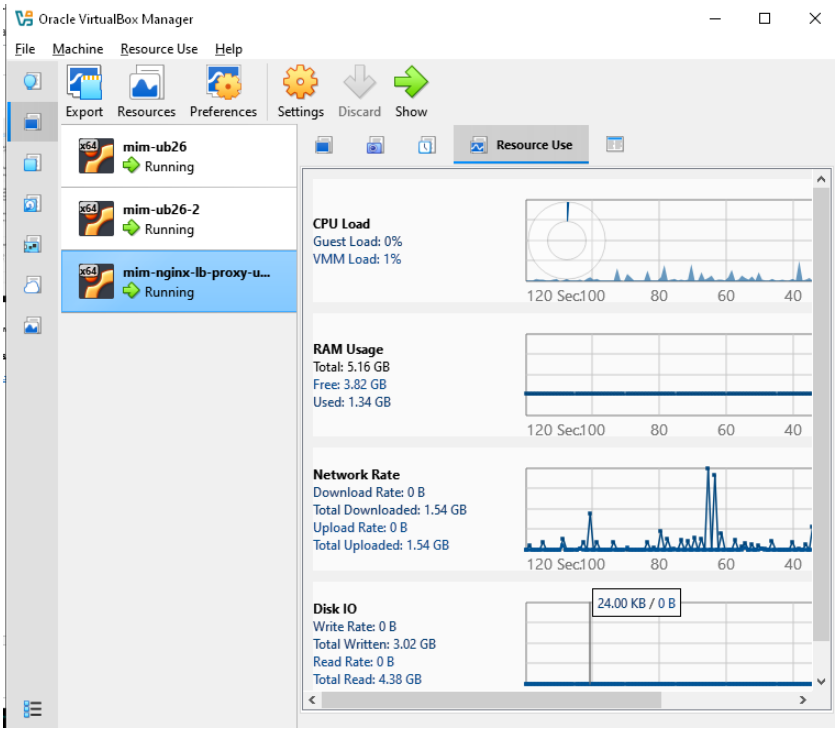
➔ VMM Interface with VMs Configured

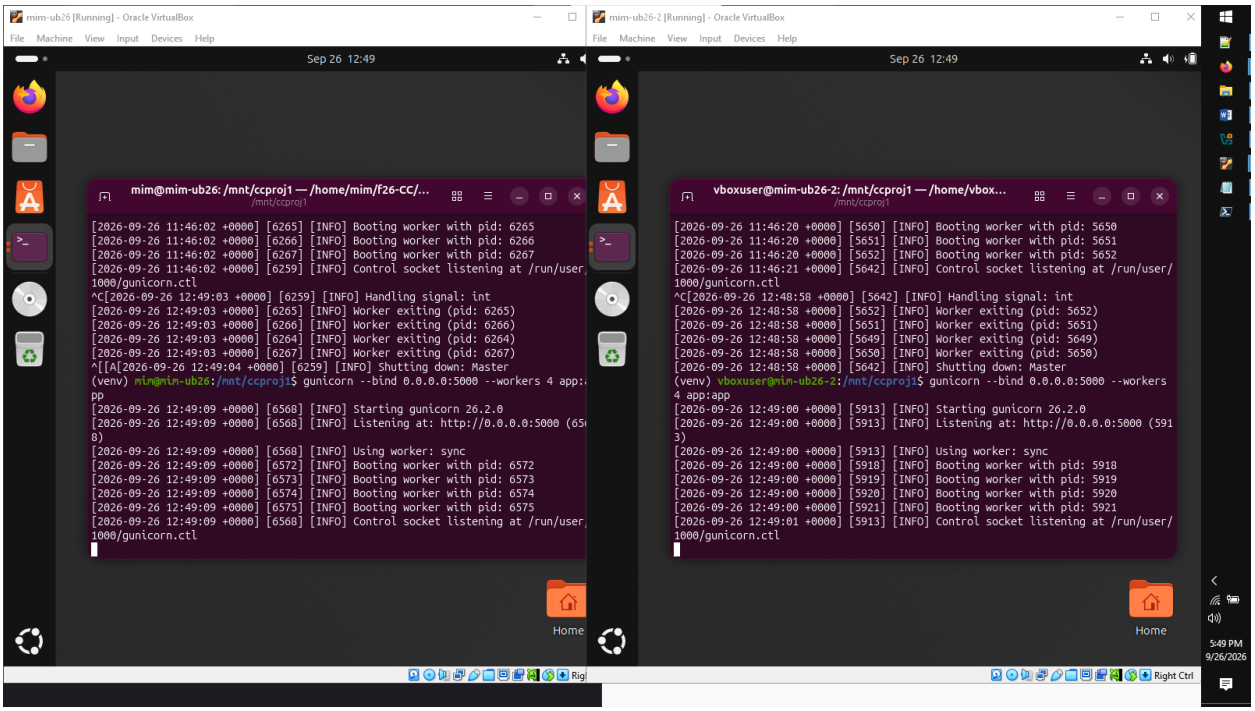


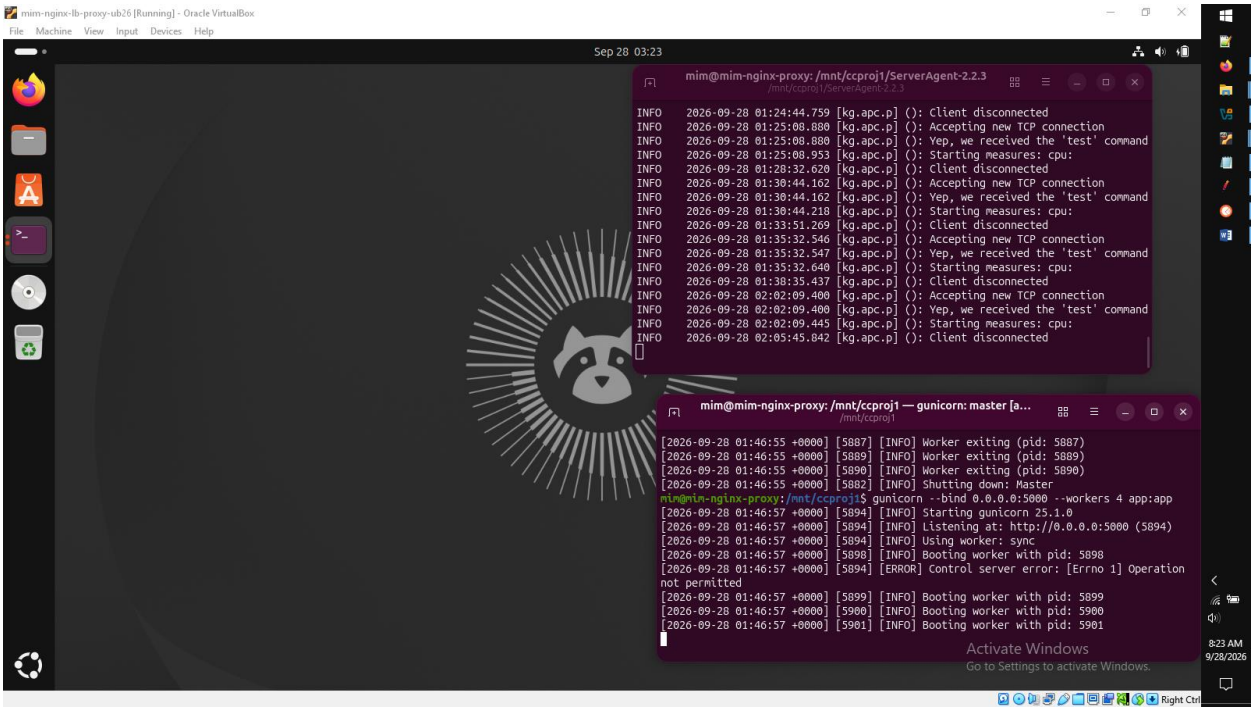


Milestone 03: Deploy Nginx as a Proxy Server in a Virtual Machine

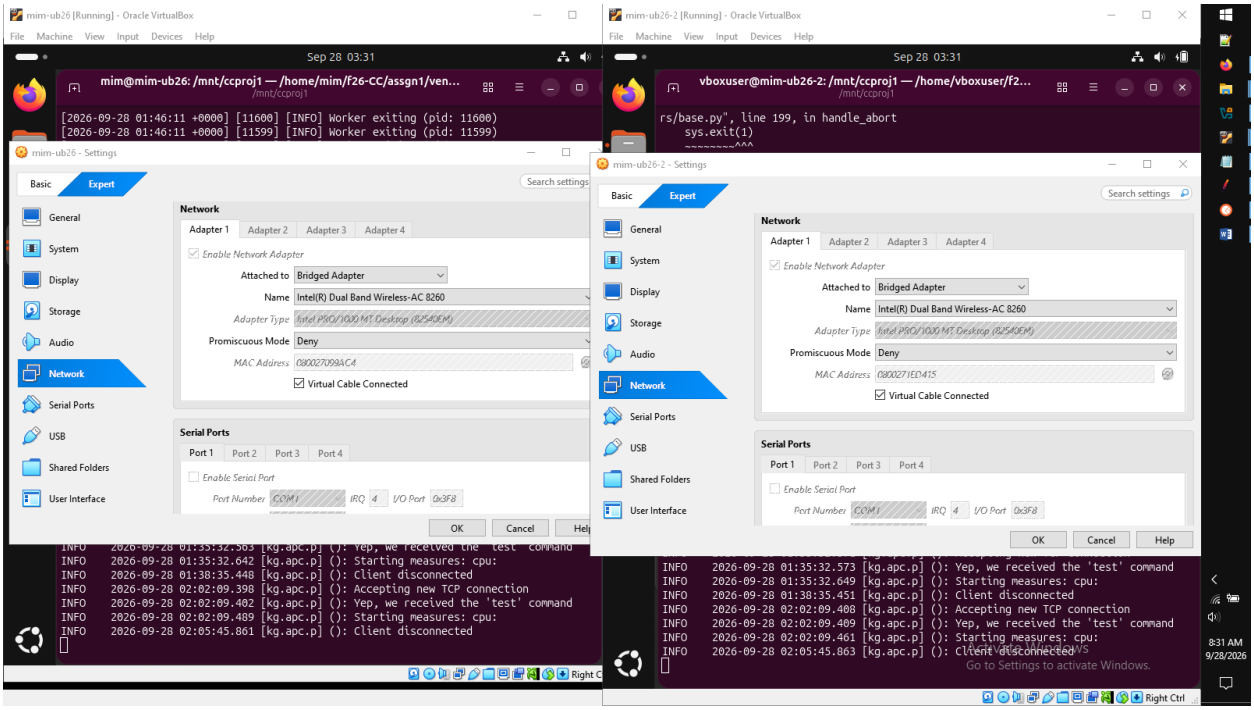
- ➔ VMs opened side by side and NGINX LB
  - VM1 IP : http://192.168.100.156:5000/ & http://192.168.100.161:5000/
  - VM2 IP : http://192.168.100.157:5000/
  - NGINX LB - VM3 IP : http://192.168.100.158:5000/

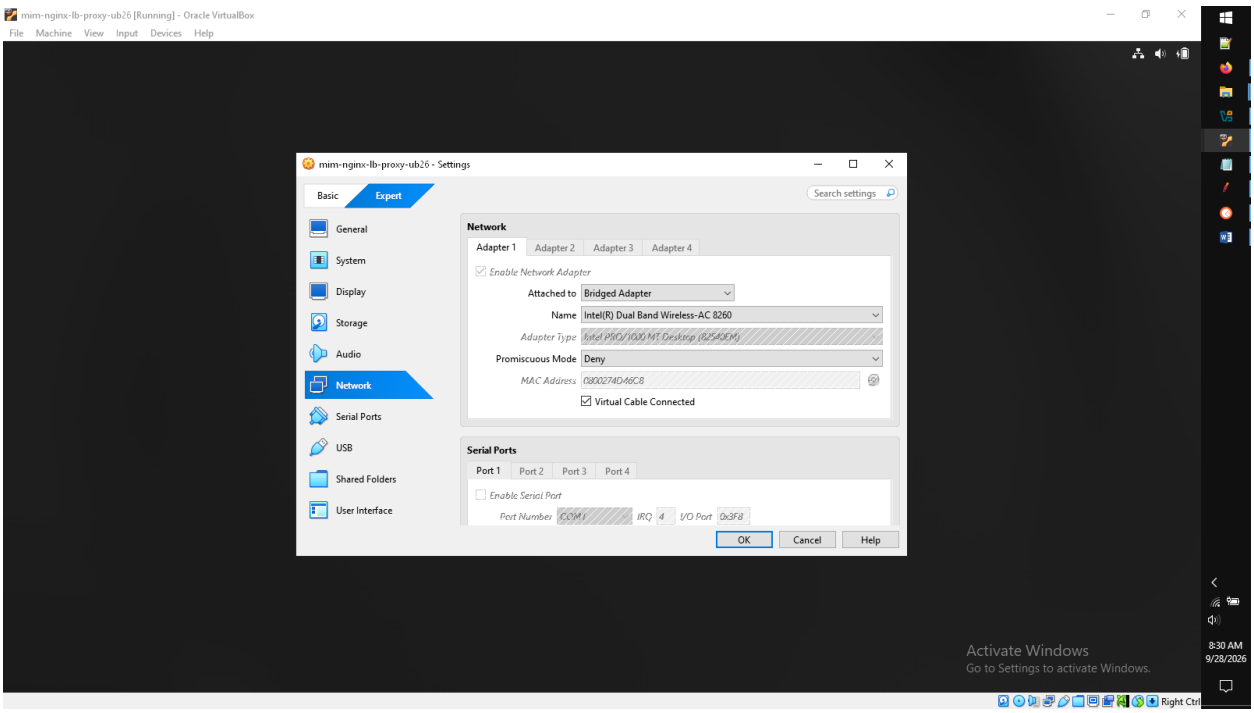



mim-ub26 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 26 12:49
mim@mim-ub26: /mnt/ccproj1 — /home/mim/f26-CC/...
/mnt/ccproj1
[2026-09-26 11:46:02 +0000] [6265] [INFO] Booting worker with pid: 6265
[2026-09-26 11:46:02 +0000] [6266] [INFO] Booting worker with pid: 6266
[2026-09-26 11:46:02 +0000] [6267] [INFO] Booting worker with pid: 6267
[2026-09-26 11:46:02 +0000] [6259] [INFO] Control socket listening at /run/user/1000/gunicorn.ctl
^C[2026-09-26 12:49:03 +0000] [6259] [INFO] Handling signal: int
[2026-09-26 12:49:03 +0000] [6265] [INFO] Worker exiting (pid: 6265)
[2026-09-26 12:49:03 +0000] [6266] [INFO] Worker exiting (pid: 6266)
[2026-09-26 12:49:03 +0000] [6264] [INFO] Worker exiting (pid: 6264)
[2026-09-26 12:49:03 +0000] [6267] [INFO] Worker exiting (pid: 6267)
^[[A[2026-09-26 12:49:04 +0000] [6259] [INFO] Shutting down: Master
(venv) mim@mim-ub26:/mnt/ccproj1$ gunicorn --bind 0.0.0.0:5000 --workers 4 app:app
[2026-09-26 12:49:09 +0000] [6568] [INFO] Starting gunicorn 26.2.0
[2026-09-26 12:49:09 +0000] [6568] [INFO] Listening at: http://0.0.0.0:5000 (6568)
[2026-09-26 12:49:09 +0000] [6568] [INFO] Using worker: sync
[2026-09-26 12:49:09 +0000] [6572] [INFO] Booting worker with pid: 6572
[2026-09-26 12:49:09 +0000] [6573] [INFO] Booting worker with pid: 6573
[2026-09-26 12:49:09 +0000] [6574] [INFO] Booting worker with pid: 6574
[2026-09-26 12:49:09 +0000] [6575] [INFO] Booting worker with pid: 6575
[2026-09-26 12:49:09 +0000] [6568] [INFO] Control socket listening at /run/user/1000/gunicorn.ctl
Home
mim-ub26-2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 26 12:49
vboxuser@mim-ub26-2: /mnt/ccproj1 — /home/vbox...
/mnt/ccproj1
[2026-09-26 11:46:20 +0000] [5650] [INFO] Booting worker with pid: 5650
[2026-09-26 11:46:20 +0000] [5651] [INFO] Booting worker with pid: 5651
[2026-09-26 11:46:20 +0000] [5652] [INFO] Booting worker with pid: 5652
[2026-09-26 11:46:21 +0000] [5642] [INFO] Control socket listening at /run/user/1000/gunicorn.ctl
^C[2026-09-26 12:48:58 +0000] [5642] [INFO] Handling signal: int
[2026-09-26 12:48:58 +0000] [5652] [INFO] Worker exiting (pid: 5652)
[2026-09-26 12:48:58 +0000] [5651] [INFO] Worker exiting (pid: 5651)
[2026-09-26 12:48:58 +0000] [5649] [INFO] Worker exiting (pid: 5649)
[2026-09-26 12:48:58 +0000] [5650] [INFO] Worker exiting (pid: 5650)
[2026-09-26 12:48:58 +0000] [5642] [INFO] Shutting down: Master
(venv) vboxuser@mim-ub26-2:/mnt/ccproj1$ gunicorn --bind 0.0.0.0:5000 --workers 4 app:app
[2026-09-26 12:49:00 +0000] [5913] [INFO] Starting gunicorn 26.2.0
[2026-09-26 12:49:00 +0000] [5913] [INFO] Listening at: http://0.0.0.0:5000 (5913)
[2026-09-26 12:49:00 +0000] [5913] [INFO] Using worker: sync
[2026-09-26 12:49:00 +0000] [5918] [INFO] Booting worker with pid: 5918
[2026-09-26 12:49:00 +0000] [5919] [INFO] Booting worker with pid: 5919
[2026-09-26 12:49:00 +0000] [5920] [INFO] Booting worker with pid: 5920
[2026-09-26 12:49:00 +0000] [5921] [INFO] Booting worker with pid: 5921
[2026-09-26 12:49:01 +0000] [5913] [INFO] Control socket listening at /run/user/1000/gunicorn.ctl
Home
5:49 PM
9/26/2026

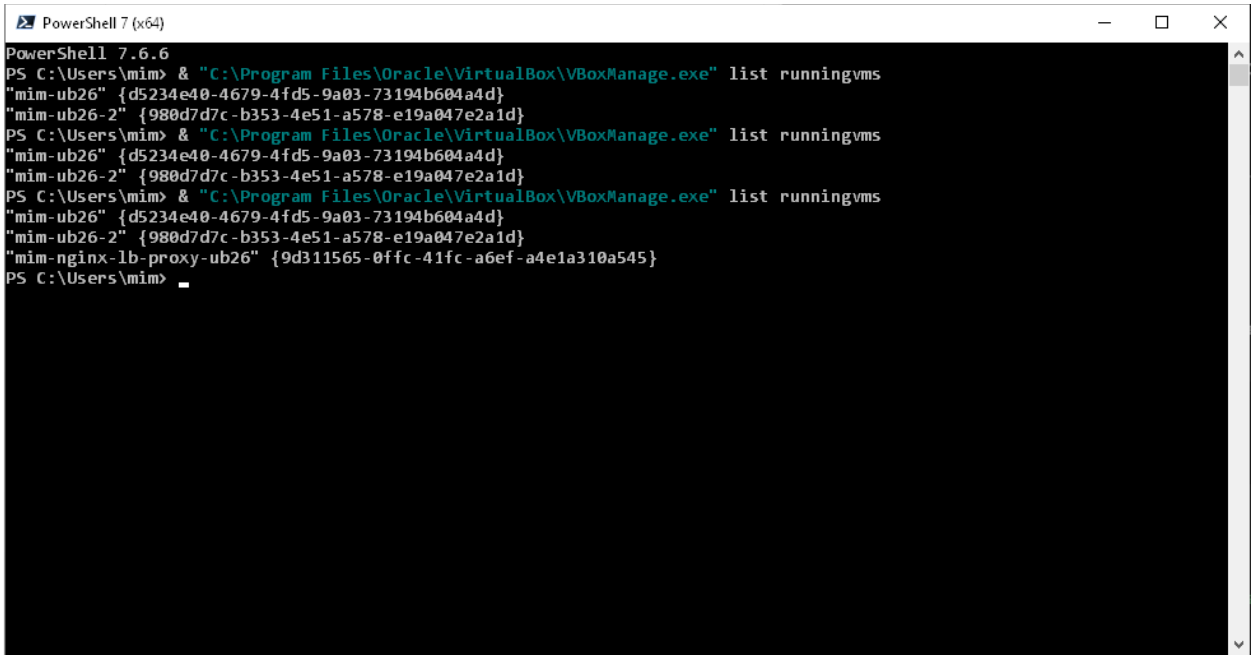

mim-nginx-lb-proxy-ub26 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 28 03:23
mim@mim-nginx-proxy: /mnt/ccproj1/ServerAgent-2.2.3
/mnt/ccproj1/ServerAgent-2.2.3
INFO 2026-09-28 01:24:44.759 [kg.apc.p] (): Client disconnected
INFO 2026-09-28 01:25:08.880 [kg.apc.p] (): Accepting new TCP connection
INFO 2026-09-28 01:25:08.880 [kg.apc.p] (): Yep, we received the 'test' command
INFO 2026-09-28 01:25:08.953 [kg.apc.p] (): Starting measures: cpu:
INFO 2026-09-28 01:28:32.620 [kg.apc.p] (): Client disconnected
INFO 2026-09-28 01:30:44.162 [kg.apc.p] (): Accepting new TCP connection
INFO 2026-09-28 01:30:44.162 [kg.apc.p] (): Yep, we received the 'test' command
INFO 2026-09-28 01:30:44.218 [kg.apc.p] (): Starting measures: cpu:
INFO 2026-09-28 01:33:51.269 [kg.apc.p] (): Client disconnected
INFO 2026-09-28 01:35:32.546 [kg.apc.p] (): Accepting new TCP connection
INFO 2026-09-28 01:35:32.547 [kg.apc.p] (): Yep, we received the 'test' command
INFO 2026-09-28 01:35:32.640 [kg.apc.p] (): Starting measures: cpu:
INFO 2026-09-28 01:38:35.437 [kg.apc.p] (): Client disconnected
INFO 2026-09-28 02:02:09.400 [kg.apc.p] (): Accepting new TCP connection
INFO 2026-09-28 02:02:09.400 [kg.apc.p] (): Yep, we received the 'test' command
INFO 2026-09-28 02:02:09.445 [kg.apc.p] (): Starting measures: cpu:
INFO 2026-09-28 02:05:45.842 [kg.apc.p] (): Client disconnected
mim@mim-nginx-proxy: /mnt/ccproj1 — gunicorn: master [a...
/mnt/ccproj1
[2026-09-28 01:46:55 +0000] [5887] [INFO] Worker exiting (pid: 5887)
[2026-09-28 01:46:55 +0000] [5889] [INFO] Worker exiting (pid: 5889)
[2026-09-28 01:46:55 +0000] [5890] [INFO] Worker exiting (pid: 5890)
[2026-09-28 01:46:55 +0000] [5882] [INFO] Shutting down: Master
mim@mim-nginx-proxy:/mnt/ccproj1$ gunicorn --bind 0.0.0.0:5000 --workers 4 app:app
[2026-09-28 01:46:57 +0000] [5894] [INFO] Starting gunicorn 25.1.0
[2026-09-28 01:46:57 +0000] [5894] [INFO] Listening at: http://0.0.0.0:5000 (5894)
[2026-09-28 01:46:57 +0000] [5894] [INFO] Using worker: sync
[2026-09-28 01:46:57 +0000] [5898] [INFO] Booting worker with pid: 5898
[2026-09-28 01:46:57 +0000] [5894] [ERROR] Control server error: [Errno 1] Operation not permitted
[2026-09-28 01:46:57 +0000] [5899] [INFO] Booting worker with pid: 5899
[2026-09-28 01:46:57 +0000] [5900] [INFO] Booting worker with pid: 5900
[2026-09-28 01:46:57 +0000] [5901] [INFO] Booting worker with pid: 5901
Activate Windows
Go to Settings to activate Windows.
8:23 AM
9/28/2026

➔ Bridged IP addresses of both application VMs

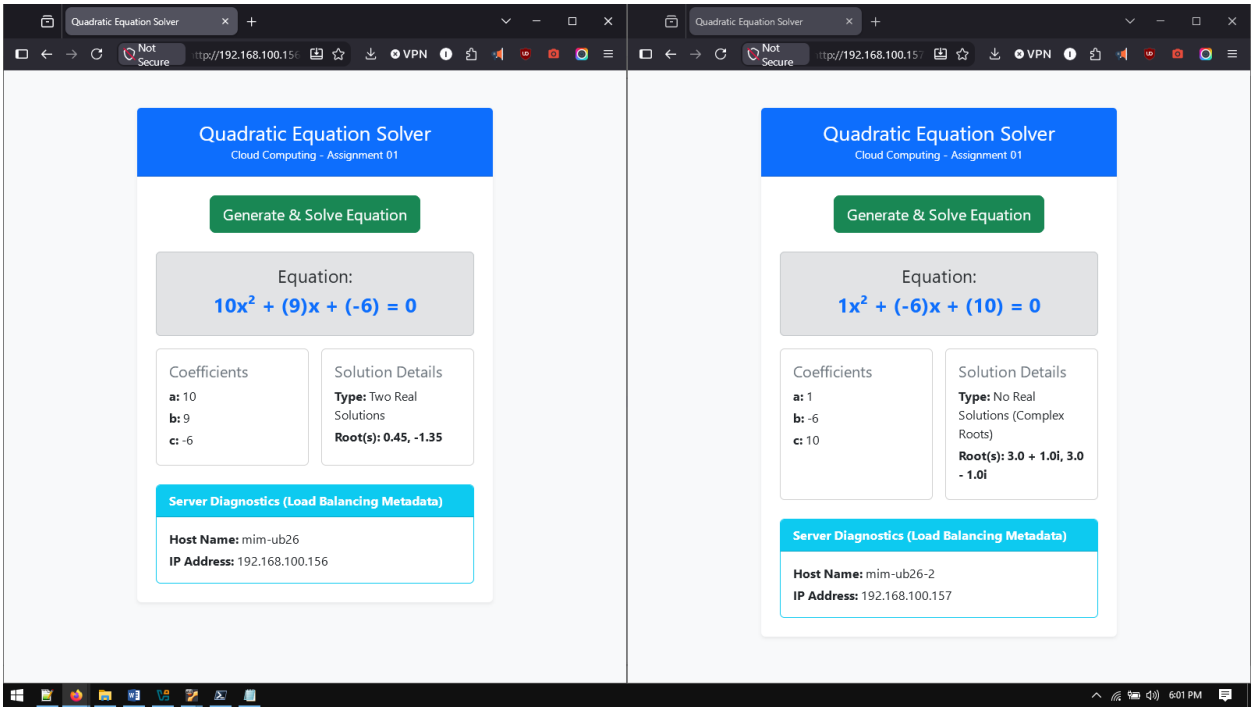




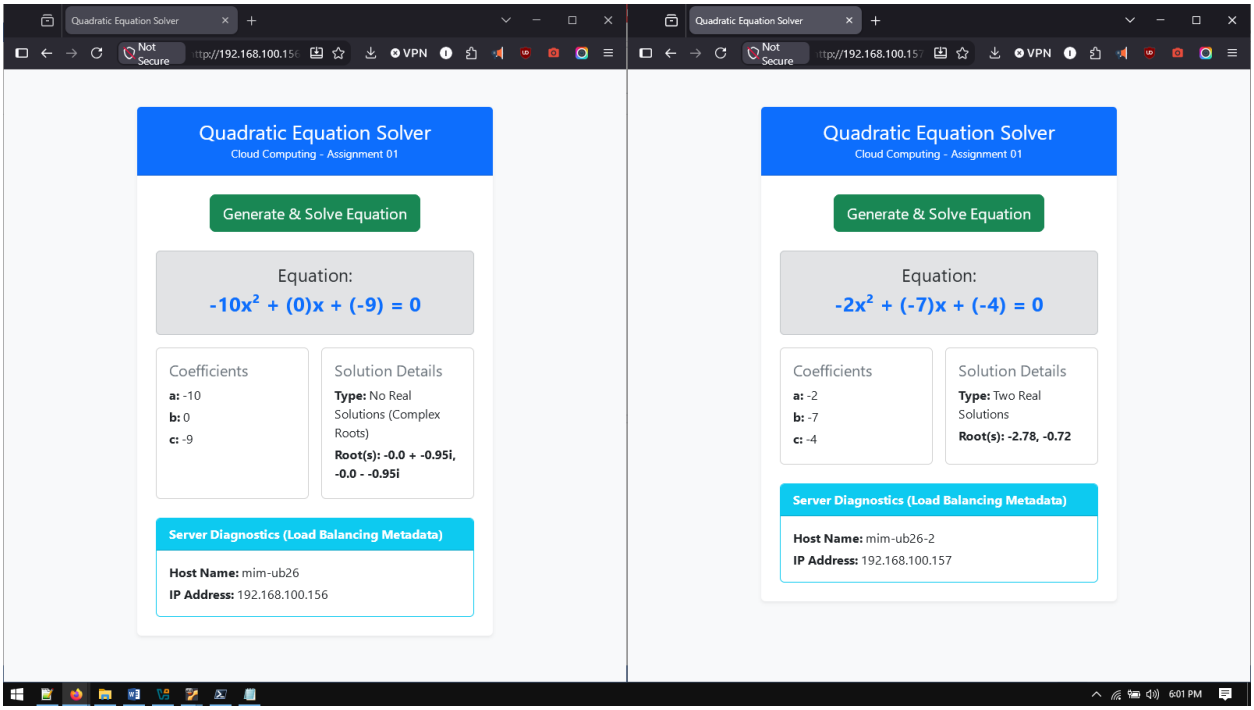
➔ Both VMs and NGINX LB are up and running



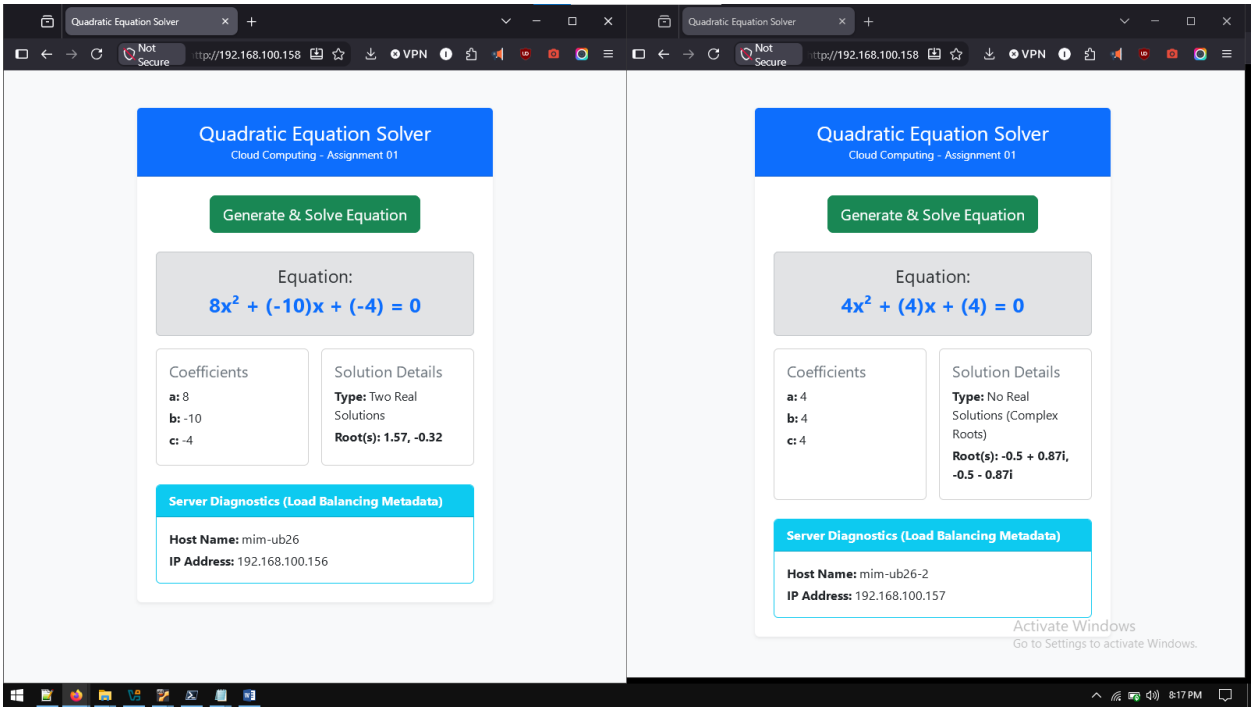
➔ Test1

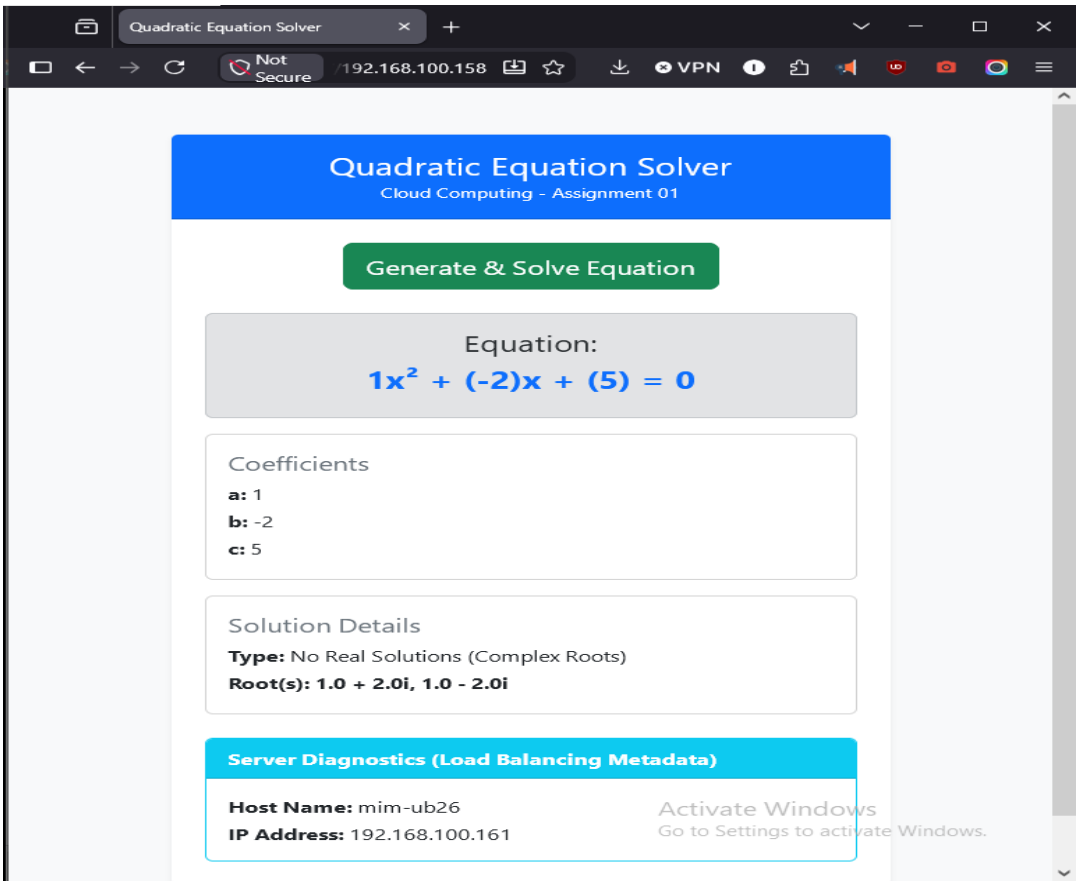


➔ Test 2

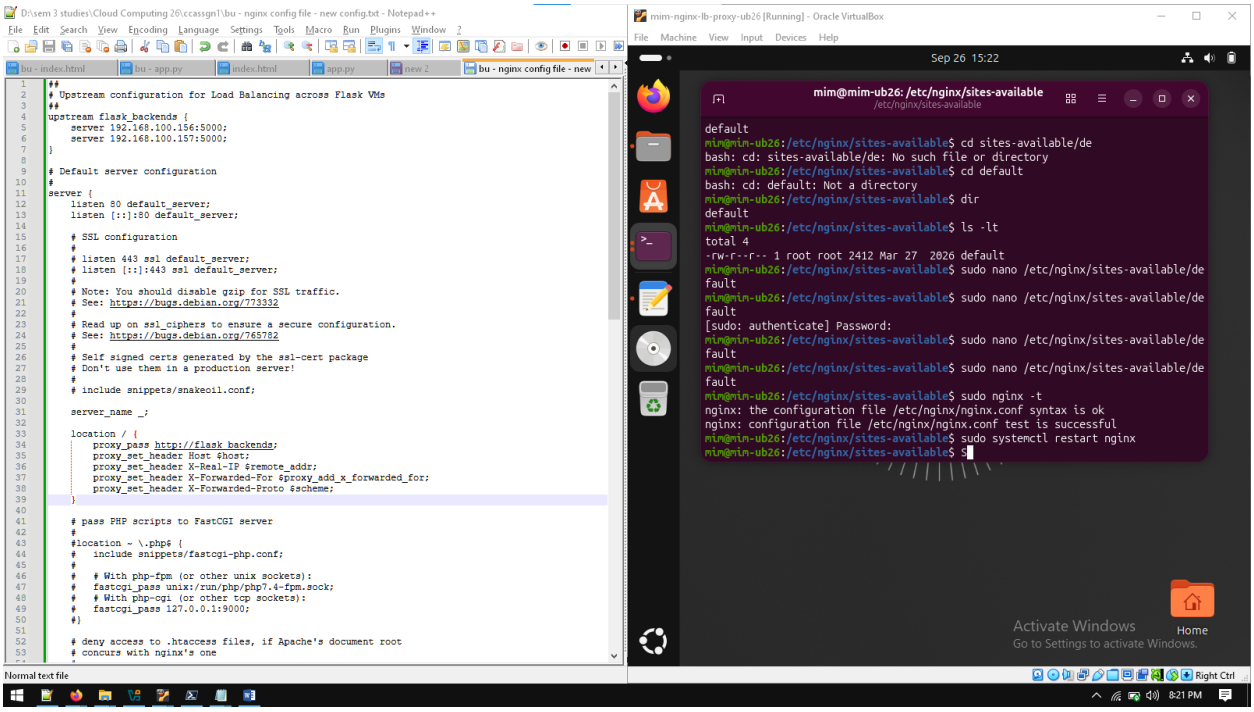


➔ NGINX (http://192.168.100.1578:5000/) sharing 1:1 load balancing





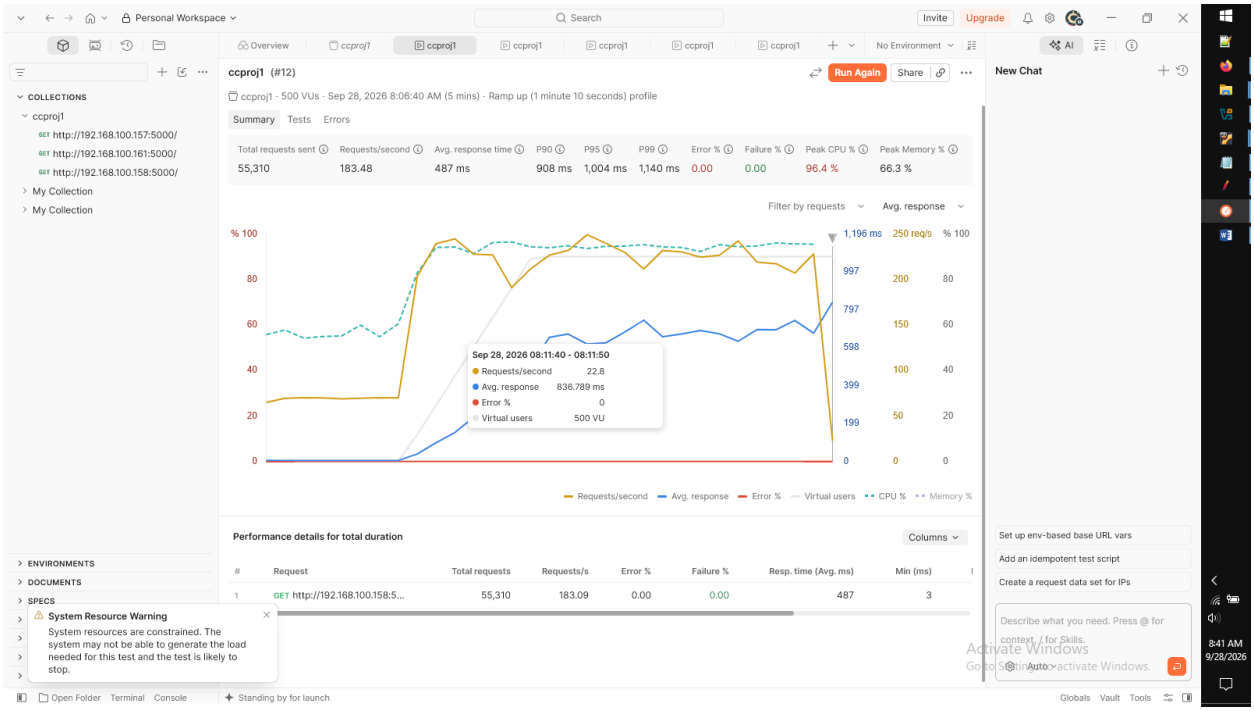
➔ NGINX LB – Config File modification, by inserting upstream for flask backend of both VMs IPs.



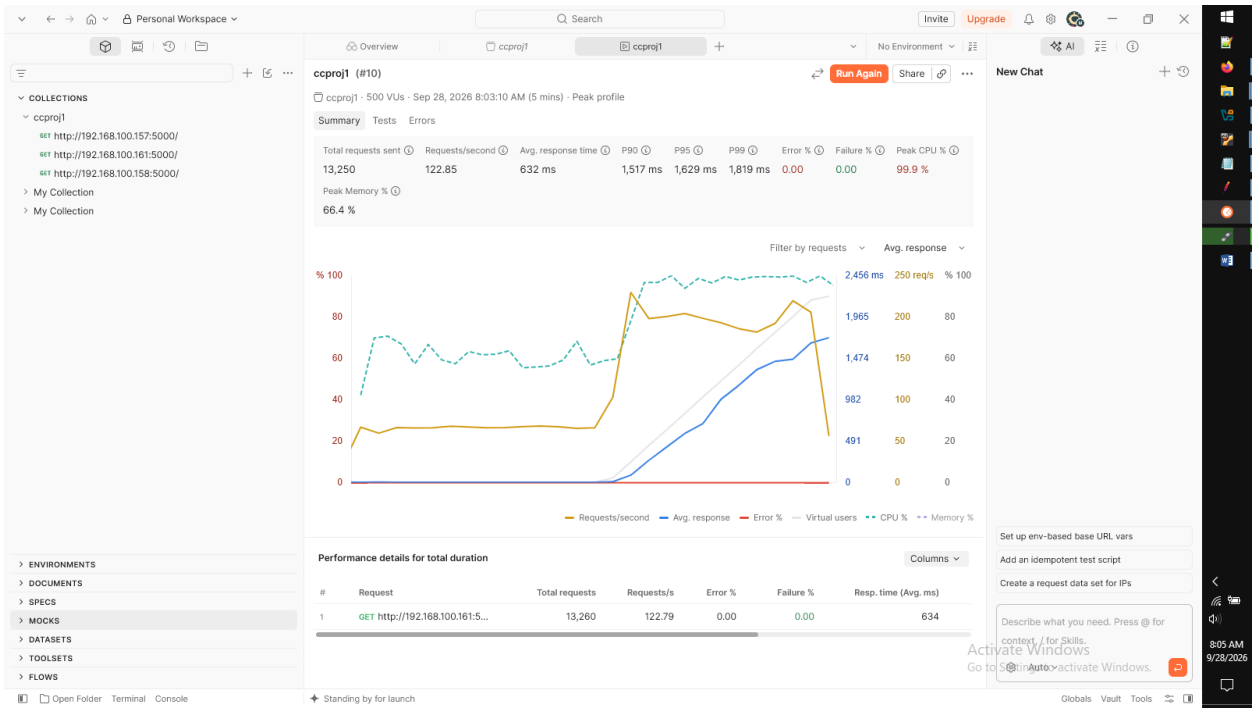
Milestone 04: Synthetic Workload Generation

➔ Postman Stress Test results

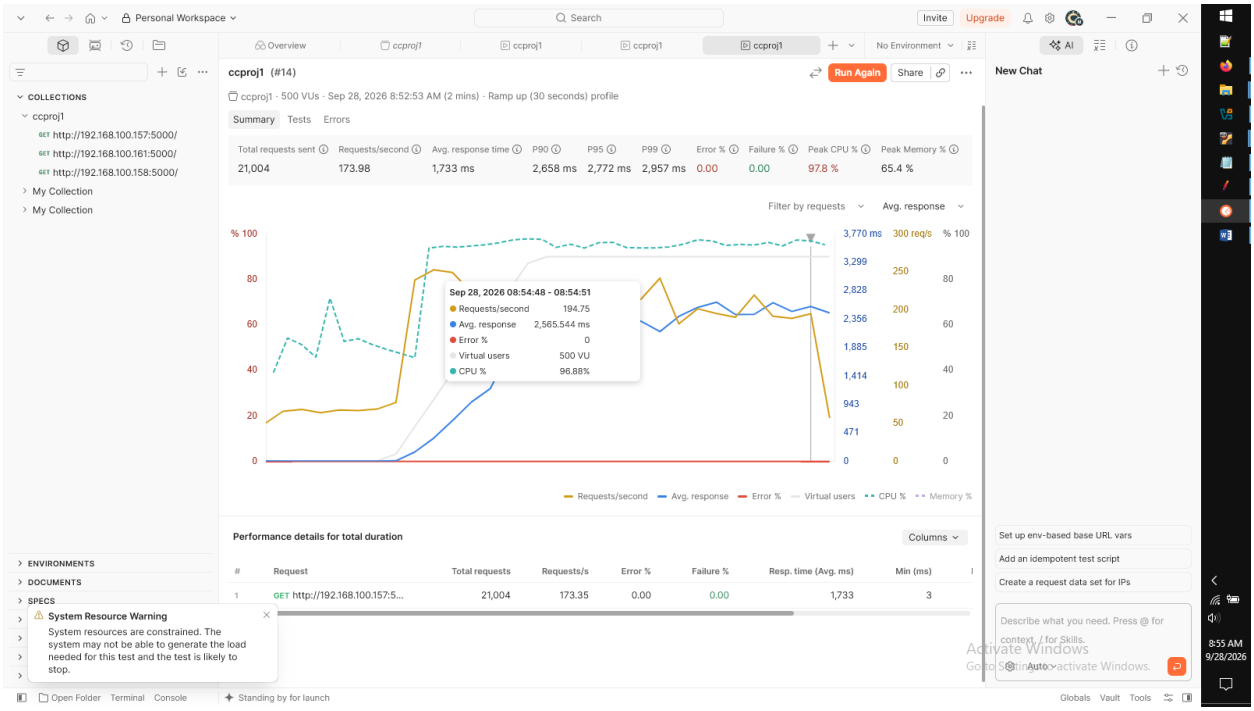
1. For http://192.168.100.158:5000/ - NGINX-LB



2. For http://192.168.100.161:5000/ - VM1



3. For http://192.168.100.157:5000/ - VM2



THE END